

Quantum Computers Explained - Limits of Human Technology

Kurzgesagt - In a Nutshell

양자 컴퓨터란 무엇인가 — 인간 기술의 한계를 넘어서

Advanced

7:17

DURATION

155

WPM

73

SEGMENTS

20

VOCABULARY

10

GRAMMAR



[10-Step Study →](#)

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01 Video Structure Overview

This 7-minute video by Kurzgesagt - In a Nutshell is structured into 7 key sections.

1 Computing's Exponential Growth Hits Physical Limits

0:00 ~ 0:32

컴퓨팅의 역사를 소개하고, 처리 능력의 기하급수적 성장이 부품이 원자 크기에 접근하면서 물리적 한계에 부딪히고 있다고 설명해.

Computer power has grown exponentially since the 1960s

Parts are approaching the size of an atom

This creates a fundamental problem for further progress

2 How Traditional Computers Work

0:39 ~ 1:50

기존 컴퓨터의 기본 구성 요소인 트랜지스터, 논리 게이트, 모듈을 설명해. 간단한 수학을 하는 7살짜리 아이들에 비유해서 단순한 연산이 어떻게 복잡한 컴퓨팅이 되는지 보여줘.

Computers consist of transistors, logic gates, and modules

Transistors are simple on/off switches processing bits

Complex calculations emerge from combining simple operations

3 The Quantum Problem: Tunneling at Atomic Scale

1:51 ~ 2:49

소형화가 양자 수준에서 왜 문제를 일으키는지 설명해. 양자 터널링으로 전자가 장벽을 통과할 수 있어서 기존 컴퓨팅이 작동하지 않게 돼. 과학자들은 양자 컴퓨터를 개발해 대응하고 있어.

At atomic scale, quantum tunneling lets electrons pass through barriers

Traditional computers stop working reliably at this scale

Scientists aim to harness quantum properties for quantum computers

4 Qubits and Superposition

2:49 ~ 4:07

큐비트와 중첩의 개념을 소개해. 고전적 비트와 달리 큐비트는 여러 상태에 동시에 존재할 수 있어서 병렬 계산이 기하급수적으로 증가할 수 있어.

Qubits can be in superposition of both 0 and 1 simultaneously

Measurement collapses the superposition to a definite state

20 qubits can store a million values in parallel

5 Entanglement and Quantum Gates

4:07 ~ 5:20

얽힘, 양자 게이트, 양자 컴퓨팅의 실제 작동 방식을 다뤄. 얽힌 큐비트는 서로 즉각적으로 반응하고, 양자 게이트는 중첩을 조작해서 대규모 병렬 계산을 수행해.

Entanglement links qubits so measuring one reveals the other's state

Quantum gates manipulate superpositions and probabilities

Superposition and entanglement enable exponentially more efficient computation

6 Applications: Databases, Encryption, and Drug Discovery

5:21 ~ 6:33

양자 컴퓨터의 실용적 응용 분야를 소개해: 그로버 알고리즘을 이용한 데이터베이스 검색, 암호화 시스템 해독, 신약 개발과 재료 과학을 위한 양자 물리학 시뮬레이션.

Database searching becomes vastly faster with quantum algorithms

Public-key encryption could be broken by quantum speed-up

Quantum simulations could revolutionize medicine through protein modeling

7 The Uncertain but Exciting Future

6:34 ~ 6:45

양자 컴퓨팅의 미래가 불확실하다는 점을 인정하며 마무리해. 전문적인 도구가 될 수도 있고 혁명이 될 수도 있으며, 탐구 정신을 강조해.

The ultimate impact of quantum computers remains unknown

They could be specialized tools or a revolution for humanity

The only way to find out is to keep pushing forward

02 Essential Vocabulary

Core words from the video that are high-value for learners.

exponentially /ˌɛkspəˈnɛnʃəli/ adverb

at a rate that becomes increasingly rapid in proportion to the growing total

기하급수적으로

Latin 'exponere' meaning 'to put out, explain' via mathematical 'exponent'

ex-

pon (put, place)

-entially

"The power of our brain machines has kept growing exponentially."

Related: exponent, exponential, exponentiation

transistor /trænˈzɪstər/ noun

a semiconductor device used to amplify or switch electronic signals and electrical power

트랜지스터 (전자 신호를 증폭하거나 전환하는 반도체 소자)

Blend of 'transfer' + 'resistor' — transfers current across a resistor

trans-

fer/resist (carry/oppose)

-or

"A transistor is the simplest form of a data processor in computers."

Related: transfer, resistor, semiconductor

quantum /ˈkwɒntəm/ adjective

relating to the branch of physics dealing with the behavior of matter and energy at the smallest scales

양자의, 양자역학에 관한

Latin 'quantum' meaning 'how much, how great'

quant (how much)

-um

"Electrons may just transfer themselves to the other side via a process called quantum tunneling."

Related: quantity, quantify, quantize

superposition /ˌsuː pərpəˈzɪʃən/ noun

a quantum state in which a particle exists in multiple states simultaneously until measured

중첩 (양자역학에서 여러 상태가 동시에 존재하는 것)

Latin 'super-' (above, on top of) + 'positio' (placing)

super-

posit (place)

-ion

"This is called superposition."

Related: position, superimpose, compose

entanglement /ɪnˈtæŋɡəlmənt/ noun

a quantum phenomenon where particles become interconnected so that the state of one instantly influences the other

얽힘, 양자 얽힘 (입자들이 상호 연결되어 한쪽의 상태가 즉시 다른 쪽에 영향을 미치는 현상)

Old French 'entangler' meaning 'to tangle up, ensnare'

en-

tangle (twist together)

-ment

"A really weird and unintuitive property qubits can have is entanglement."

Related: entangle, tangle, disentangle

polarization /ˌpɒləraɪˈzeɪʃən/ noun

the orientation of vibrations of a wave, especially light, in a particular direction

편광 (빛 등의 파동이 특정 방향으로 진동하는 것)

French 'polarisation' from 'polaire' (polar)

polar (of the pole)

-ization

"Zero and one are this system's possible states, like the photon's horizontal or vertical polarization."

Related: polarize, polar, bipolar

qubit /ˈkjuː bɪt/ noun

the basic unit of quantum information, capable of existing in superposition of states

큐비트 (양자 컴퓨터의 기본 정보 단위)

Portmanteau of 'quantum' + 'bit'

qu-

bit (binary digit)

"Quantum computers use qubits, which can also be set to one of two values."

Related: quantum, bit, byte

encryption /ɪnˈkrɪptʃən/ noun

the process of converting information into a secret code to prevent unauthorized access

암호화

Greek 'en-' (in) + 'kryptein' (to hide)

en-

crypt (hide)

-ion

"Your data is being kept secure by an encryption system."

Related: encrypt, decrypt, cryptography

simulation /ˌsɪmjʊˈleɪʃən/

noun

an imitation of a real-world process or system using a computer model

시뮬레이션, 모의 실험

Latin 'simulare' meaning 'to imitate, represent'

simul (imitate)

-ation

"Another really exciting new use is simulations."

Related: simulate, similar, simultaneous

tunneling /ˈtʌnəlɪŋ/

noun

a quantum mechanical phenomenon where a particle passes through a potential energy barrier

터널링 (입자가 에너지 장벽을 통과하는 양자역학적 현상)

Old French 'tonel' (cask, barrel) — originally referring to a passage through something

tunnel (underground passage)

-ing

"Electrons may just transfer themselves to the other side via a process called quantum tunneling."

Related: tunnel, channel, passage

nanometer /ˈnænəˌmiː tər/

noun

a unit of length equal to one billionth of a meter

나노미터 (10억분의 1미터)

Greek 'nanos' (dwarf) + 'metron' (measure)

nano-

meter (measure)

"A typical scale for transistors is 40 nanometers."

Related: nanotechnology, micrometer, kilometer

deduce /dɪˈdjuː s/

verb

to arrive at a conclusion by reasoning from given information

추론하다, 연역하다

Latin 'deducere' meaning 'to lead down, derive'

de-

duc (lead)

-e

"You can directly deduce properties of its partners without having to look."

Related: deduction, deduce, induce, reduce

instantaneously /ˌɪnstənˈteɪniəsli/ adverb

happening or done in an instant; immediately

즉각적으로, 순간적으로

Latin 'instantaneus' from 'instare' (to stand upon, be present)

in-

stant (stand)

-aneously

"Each of the qubits react to a change in the other's state instantaneously."

Related: instant, instantaneous, instance

configuration /kənˌfɪɡjʊ'reɪʃən/ noun

an arrangement of elements in a particular form or combination

구성, 배치, 설정

Latin 'configurare' meaning 'to form, shape after'

con-

figur (shape, form)

-ation

"Four classical bits can be in one of two to the power of four different configurations at a time."

Related: configure, figure, figurative

algorithm /ˈælgərɪðəm/ noun

a process or set of rules to be followed in calculations or problem-solving operations

알고리즘 (문제 해결을 위한 절차나 규칙의 집합)

Named after al-Khwarizmi, a 9th-century Persian mathematician

algorithm (from al-Khwarizmi)

"Quantum algorithms need only the square root of that time."

Related: algorithmic, logarithm, computation

molecule /ˈmɒlɪkjʊl/ noun

a group of atoms bonded together, representing the smallest unit of a chemical compound

분자

French 'molécule' from Latin 'moles' (mass) + diminutive '-cula'

mol (mass)

-ecule (small)

"Even for bigger structures such as molecules, they often lack accuracy."

Related: molecular, mole, atom

revolutionize /ˌrɛvəˈluːʃənəɪz/ verb

to change something radically or fundamentally

혁명적으로 변화시키다

Latin 'revolutio' meaning 'a turn around' from 'revolvere' (to turn back)

re-

volut (turn)

-ionize

"Quantum simulations could provide new insights on proteins that might revolutionize medicine."

Related: revolution, revolve, evolve

photon /ˈfɒtɒn/ noun

a particle representing a quantum of light or other electromagnetic radiation

광자 (빛이나 전자기 복사의 양자 단위)

Greek 'phos/phot-' meaning 'light'

phot (light)

-on (particle)

"A qubit can be any two-level quantum system, such as a spin in a magnetic field or a single photon."

Related: photograph, photosynthesis, electron

03 Additional Vocabulary

Supplementary words that appear in the video and are useful for expanding your vocabulary.

unintuitive /ˌʌnɪnˈtjuː ɪtɪv/ adjective

not easy to understand or grasp by instinct; counterintuitive

직관에 반하는, 직감적으로 이해하기 어려운

Latin 'un-' (not) + 'intueri' (to look at, consider)

un-

intuit (to perceive directly)

-ive

"A really weird and unintuitive property qubits can have is entanglement."

Related: intuitive, intuition, counterintuitive

vastly /ˈvɑː stli/ adverb

to a very great extent; immensely

엄청나게, 대단히

Latin 'vastus' meaning 'immense, huge, empty'

vast (huge)

-ly

"In some areas they are vastly superior."

Related: vast, vastness, immensely

04 Grammar Patterns

Key grammar structures used in the video. Each pattern includes real examples.

Present perfect continuous: has/have + been + verb-ing

Used to describe an action that started in the past and continues to the present with emphasis on duration. 'Has kept growing' shows the ongoing nature of technological progress.

현재완료 진행형은 과거에 시작된 행동이 현재까지 계속되고 있음을 강조할 때 사용해. 'has kept growing'은 기술 발전이 계속 진행 중임을 보여줘.

The power of our brain machines has kept growing exponentially.

Be about to + verb (imminent future)

Used to express that something is on the verge of happening. It creates a sense of urgency or immediacy about an upcoming event.

'be about to + 동사'는 어떤 일이 곧 일어나려 한다는 것을 표현할 때 사용해. 긴박감이나 즉각성을 줘. '막 ~하려 하다'라는 의미야.

But this process is about to meet its physical limits.

Be made up of + noun (composition)

Used to describe what something consists of or is composed of. It is a common way to explain the parts that form a whole.

'be made up of'는 어떤 것이 무엇으로 구성되어 있는지 설명할 때 사용해. '~로 이루어져 있다'라는 의미야.

A computer is made up of very simple components doing very simple things.

This information is made up of bits.

Once + clause, clause (conditional sequence)

Used to show that one event triggers or enables another. It implies that the first action is a prerequisite for the second.

'Once + 절'은 한 가지 일이 다른 일을 가능하게 하거나 촉발함을 보여줘. 첫 번째 행동이 두 번째의 전제 조건임을 암시해. '일단 ~하면'이라는 뜻이야.

Once you can add, you can also multiply, and once you can multiply, you can basically do anything.

May/might + verb (possibility/speculation)

Used to express that something is possible but not certain. In scientific contexts, it indicates theoretical possibility or uncertainty.

'may/might + 동사'는 가능하지만 확실하지 않은 일을 표현할 때 사용해. 과학적 맥락에서는 이론적 가능성이나 불확실성을 나타내.

Electrons may just transfer themselves to the other side of a blocked passage.

Quantum simulations could provide new insights on proteins that might revolutionize medicine.

As long as + clause (condition duration)

Used to set a condition that must remain true for the main clause to be valid. It means 'during the entire time that' or 'provided that.'

'as long as + 절'은 주절이 유효하기 위해 유지되어야 하는 조건을 설정할 때 사용해. '~하는 한', '~하는 동안'이라는 뜻이야.

So, as long as it's unobserved, the qubit is in a superposition of probabilities for zero and one.

What this means is that + clause (reformulation)

Used to rephrase or clarify a complex idea in simpler terms. It signals an explanation of the practical implications of what was just described.

'What this means is that'은 복잡한 아이디어를 더 쉽게 바꿔 설명할 때 사용해. 방금 설명한 것의 실질적인 의미를 풀어서 말하는 신호야.

What this means is that you get the entire lot of calculations that are possible with your setup all done at the same time.

So while + clause, clause (concession)

Used to acknowledge one point while introducing a contrasting conclusion. It balances two ideas, recognizing a limitation before stating a strength.

'So while + 절'은 한 가지 점을 인정하면서 대조적인 결론을 소개할 때 사용해. '~이긴 하지만, ~이다'라는 의미로 두 아이디어의 균형을 잡아.

So while quantum computers will probably not replace our home computers, in some areas they are vastly superior.

By + verb-ing (means/method)

Used to explain the method or means by which something is achieved. It answers the 'how' of an action.

'by + 동명사(-ing)'는 어떤 것을 달성하는 방법이나 수단을 설명할 때 사용해. 행동의 '어떻게'에 답하는 표현이야.

But by cleverly exploiting superposition and entanglement, this can be exponentially more efficient.

Scientists are trying to use these unusual quantum properties to their advantage by building quantum computers.

Why not + verb? (rhetorical suggestion)

Used to propose an idea as if it is an obvious or natural solution. It is a rhetorical way to introduce an innovative approach.

'Why not + 동사?'는 어떤 아이디어를 당연하거나 자연스러운 해결책으로 제안할 때 사용해. 혁신적인 접근 방식을 수사적으로 소개하는 방법이야.

So why not simulate quantum physics with actual quantum physics?

05 Connected Speech & Pronunciation

Natural pronunciation patterns found in the talk. Practice these to sound more natural!

Linking (연음)

made up of → **메이더퍼브** /meɪdʌpəv/
'made'의 /d/가 'up'의 모음 /ʌ/와 연결되고, 'up'의 /p/가 약화된 'of' /əv/와 이어져서 세 단어가 하나의 덩어리처럼 흘러가요.
'메이드 업 오브'라고 따로 끊지 말고, '메이더퍼브'처럼 한 덩어리로 이어서 발음해 보세요.

measure it → **메저릿** /ˈmeɪʒərɪt/
'measure'의 마지막 소리가 'it'의 /ɪ/와 자연스럽게 연결되어 전체 구가 한 단어처럼 들려요.
'메저 잇'이라고 끊지 말고, '메저릿'처럼 /r/ 소리를 'it'과 이어서 발음해 보세요.

do it in a → **두이리너** /duː ɪtnə/
여러 자음-모음 연결이 연쇄적으로 이어져요: 'it'의 /t/가 'in'의 /ɪ/와, 'in'의 /n/이 'a'의 /ə/와 연결되어 전체 구가 하나의 연속된 소리로 흘러가요.
'두 잇 인 어'라고 끊지 말고, '두이리너'처럼 자음과 모음을 연결하면서 한 호흡으로 발음해 보세요.

Reduction (축약)

set to either → **세터 이더** /sɛtə ˈiː ðər/
'to'가 /tə/로 약화되고, 'set'의 /t/와 바로 연결돼요. 'either'도 첫 모음이 약해져서 전체 구가 매끄럽게 들려요.
'셋 투 이더'가 아니라 '세터 이더'처럼 'to'를 약하게 /tə/로 발음하며 앞 단어와 연결해 보세요.

can be → **컨비** /kənbi/
'can'이 강세를 받지 않는 위치에서 약형 /kən/으로 축약돼요. 모음 /æ/가 슈와 /ə/로 변해서 구가 빠르게 흘러가요.
'캔 비'라고 강하게 발음하지 말고, '컨비'처럼 'can'을 약하게 발음해 보세요. 부정문 'can't'와 구별하는 중요한 포인트예요.

Elision (탈락)

just stop → **저스 스탑** /dʒʌs stɒp/
'just'의 마지막 /t/가 'stop'의 /s/ 앞에서 탈락해요. 파열음 뒤에 다른 자음이 오면 첫 번째 파열음이 생략되는 경우가 많아요.
'저스트 스탑'이라고 /t/를 분명히 발음하지 말고, '저스 스탑'처럼 /t/를 생략해 보세요.

most famous → **모우스 페이머스**

/moʊs 'feɪməs/

'most'의 마지막 /t/가 'famous'의 /f/ 앞에서 탈락해요. 영어에서 흔히 일어나는 자음군 단순화 현상이에요.

'모우스트 페이머스'가 아니라 '모우스 페이머스'처럼 /t/를 생략하면 더 자연스러워요.

Assimilation (동화)

in parallel → **임 패럴렐**

/ɪm 'pærələl/

'in'의 /n/이 'parallel'의 양순음 /p/ 앞에서 /m/으로 동화돼요. 비음이 다음 자음의 조음 위치에 맞춰 변하는 조음위치 동화 현상이에요.

'인 패럴렐'이 아니라 '임 패럴렐'처럼 /n/을 /m/으로 바꿔서 발음해 보세요. 입술을 다물면서 자연스럽게 이어져요.

06 Signal Expressions

Key phrases used to structure the content. Learning these helps you follow presentations.

TRANSITION

"For most of our history"

역사적 맥락을 설정하며 이야기를 시작해요

EMPHASIS

"But this process is about to meet its physical limits"

핵심 문제를 예고하며 긴박감을 조성해요

TRANSITION

"To understand why this is a problem, we have to clear up some basics"

기본 개념 설명으로 전환하며 청중을 안내해요

TRANSITION

"However, with parts getting tinier and tinier"

양자 문제로의 전환을 알리는 역접 표현이에요

SUMMARY

"In a nutshell"

복잡한 개념을 간결하게 요약하겠다는 신호예요

TRANSITION

"To solve this problem, scientists are trying to"

문제에서 해결책으로 전환하며 양자 컴퓨터를 소개해요

EMPHASIS

"This is called superposition"

핵심 용어를 정의하며 개념을 명확하게 해요

EMPHASIS

"Superposition is a game-changer"

중첩의 중요성을 강조하며 실질적 의미를 설명하기 시작해요

SUMMARY

"What this means is that"

복잡한 설명을 실질적인 의미로 바꿔 설명해요

TRANSITION

"So while quantum computers will probably not replace our home computers"

이론에서 실제 응용 분야로 전환하며 기대를 조정해요

TRANSITION

"Another really exciting new use is"

새로운 응용 분야로 자연스럽게 전환해요

SUMMARY

"We have no idea where the limits of technology are"

전체 영상을 마무리하며 열린 결론으로 끝내요

Keep Learning!

Generated by stdyLang — AI-Powered Language Learning
[youtube.com/watch?v=JhHMJCUmQ28](https://www.youtube.com/watch?v=JhHMJCUmQ28)